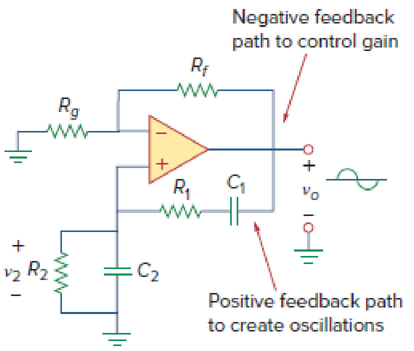
Chapter 10, Problem 16PP

Problem

In the Wien-bridge oscillator circuit in Fig. 10.42, let $R1 = R2 = 2.5\ \text{k}\Omega$, $C1 = C2 = 1$ nF. Determine the frequency *fo* of the oscillator.

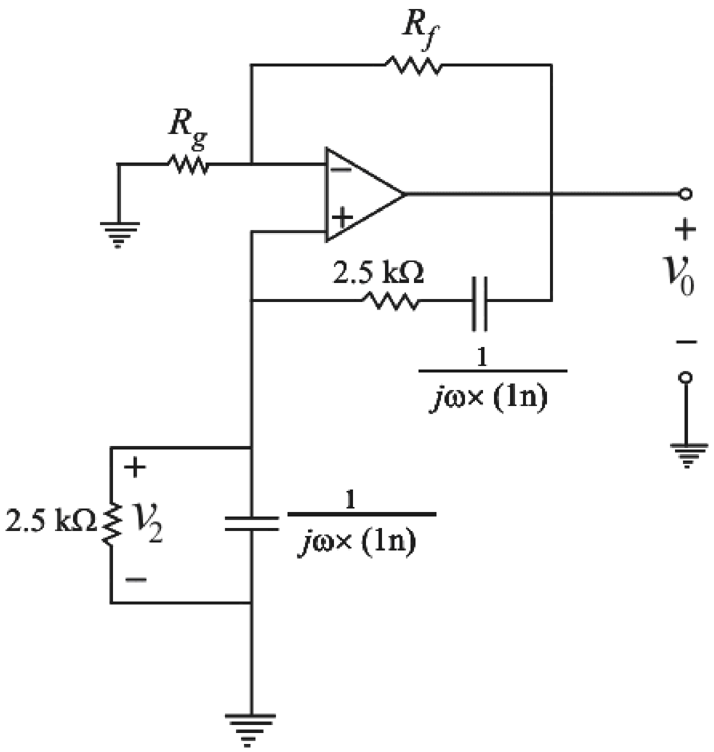
Figure 10.42: Wien-bridge oscillator.



Step-by-step solution

Step 1 of 5

Given circuit can be represented as,



Step 2 of 5

By using voltage division to above circuit,

$$v_2 = v_0 \times \left[\frac{\left((2.5\times10^3) \| \frac{1}{j\omega\times(1\times10^{-9})} \right)}{(2.5\times10^3) \| \left(\frac{1}{j\omega\times(1\times10^{-9})} \right) + \left[(2.5\times10^3) + \frac{1}{j\omega\times(1\times10^{-9})} \right]} \right]$$

$$v_2 = v_0 \left[\frac{\dfrac{\left(\dfrac{2.5\times10^3}{j\omega\times(1\times10^{-9})} \right)}{2.5\times10^3 + \dfrac{1}{j\omega\times(1\times10^{-9})}}}{\left(\dfrac{\dfrac{2.5\times10^3}{j\omega\times(1\times10^{-9})}}{2.5\times10^3 + \dfrac{1}{j\omega\times(1\times10^{-9})}} \right) + \left[(2.5\times10^3) + \dfrac{1}{j\omega\times(1\times10^{-9})} \right]} \right]$$

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$$\frac{v_2}{v_0} = \left[\frac{\left(\dfrac{2.5\times10^3}{(2.5\times10^3)\times(j\omega\times1\times10^{-9})+1} \right)}{\left(\dfrac{2.5\times10^3}{(2.5\times10^3)\times(j\omega\times1\times10^{-9})+1} \right) + \dfrac{(2.5\times10^3)\times(j\omega\times10^{-9})+1}{(j\omega\times1\times10^{-9})}} \right]$$

$$\frac{v_2}{v_0} = \left[\frac{2.5\times10^3}{(2.5\times10^3)\times(j\omega\times1\times10^{-9}) + \left[(2.5\times10^3)\times(j\omega\times10^{-9})+1 \right] \times \left[(2.5\times10^3)\times(j\omega\times1\times10^{-9})+1 \right]} \right]$$

$$\frac{v_2}{v_0} = \left[\frac{(2.5\times10^3)\times(j\omega\times1\times10^{-9})}{\left[(2.5\times10^3)\times(j\omega\times1\times10^{-9}) \right] + \left[(2.5\times10^3)\times(j\omega\times10^{-9})+1 \right]^2} \right]$$

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$$\frac{v_2}{v_0} = \left[\frac{(2.5\times10^3)\times(j\omega\times1\times10^{-9})}{\left[(2.5\times10^3)\times(j\omega\times1\times10^{-9}) \right] + \left\{ \begin{array}{l} \left[(2.5\times10^3)\times(j\omega\times1\times10^{-9}) \right]^2 \\ + \left[2(2.5\times10^3)\times(j\omega\times1\times10^{-9}) \right] + 1 \end{array} \right\}} \right]$$

$$\frac{v_2}{v_0} = \left[\frac{(2.5\times10^3)\times(\omega\times1\times10^{-9})}{\left[3\times(2.5\times10^3)\times(\omega\times1\times10^{-9}) \right] + \left[(2.5\times10^3)^2 \times \left(j\omega^2 \times (1\times10^{-9})^2 \right) \right] + \frac{1}{j}} \right]$$

$$\therefore \frac{v_2}{v_0} = \left[\frac{(2.5\times10^3)(\omega\times1\times10^{-9})}{\left[3\times(2.5\times10^3)\times(\omega\times1\times10^{-9}) \right] + j\left[\left(\omega^2 \times (2.5\times10^3)^2 \times (1\times10^{-9})^2 \right) - 1 \right]} \right]$$

Step 5 of 5

From the above equation,

v_2 must be in phase with v_0, which implies that the ratio must be purely real. Hence, the imaginary part must be zero. Setting the imaging part equal to zero gives the oscillation frequency ω_0 as

$$\left[\omega_0^2 \times \left(2.5\times10^3\right)^2 \times \left(1\times10^{-9}\right)^2\right] - 1 = 0$$

$$\omega_0^2 \times \left(2.5\times10^3\right)^2 \times \left(1\times10^{-9}\right)^2 = 1$$

$$\omega_0^2 = \frac{1}{\left(2.5\times10^3\right)^2 \times \left(1\times10^{-9}\right)^2}$$

$$\omega_0 = \frac{1}{\sqrt{\left(2.5\times10^3\right)^2 \times \left(1\times10^{-9}\right)^2}}$$

$$2\pi f_0 = \frac{1}{\left(2.5\times10^3\right) \times \left(1\times10^{-9}\right)}$$

$$2\pi f_0 = 400\times10^3$$

$$f_0 = \frac{400\times10^3}{2\pi}$$

$$f_0 = \frac{400\times10^3}{2\times3.1416}$$

$$\therefore \boxed{f_0 = 63.66\,\text{k Hz}}$$